Anomaly Detection through on-line Isolation Forest: an Application to Plasma Etching

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***Abstract*—Advanced Monitoring Systems are fundamental in advanced manufacturing for control, quality and maintenance purposes. Nowadays, with the increasing availability of data in production and equipment, the need for high-dimensional Anomaly Detection techniques is thriving; anomalies are data patterns that have different data characteristics from normal production instances and that may be associated with faults or drifts in production. Tools for dealing with high-dimensional monitoring problems are provided by Machine Learning: in this paper, we test the performance of a state-of-the-art anomaly detection technique, called Isolation Forest, on a real industrial dataset related to Etching, one of the most important semiconductor manufacturing process. The monitoring has been performed exploiting Optical Spectroscopy Data.**

***Index Terms*—Anomaly Detection, Advanced Monitoring Systems, Data Mining, Etching, Fault-diagnosis, Isolation Forest, Machine Learning, Optical Emission Spectroscopy, Semiconductor Manufacturing**

I. Introduction

Smart production monitoring is a fundamental activity in semiconductor manufacturing for quality [1], [2], control [3]–[5] and maintenance [6] purposes. Advanced Monitoring Systems (AMSs) aim at detecting anomalies and trends; anomalies are data patterns that have different data characteristics from normal instances [7], while trends are tendencies of production to move in a particular direction over time. A statistical definition of anomaly is *outlier*, an observation that is distant from other observations; on the other hand, non-anomaly observations are referred as *inliers*.

A classical approach to production monitoring is the employment of univariate Control Charts [8], where the crossing of predefined control limits trigger alerts/warnings. However, one of the main criticisms to a monitoring of a process only thorough Control Charts is that multivariate aspects are not captured by this approach; a 2-dimensional example of this limitation is represented in Fig. 1: an outlier is within the control limits (Upper Control Limit - UCL - and Lower Control Limit - LCL) are goes undetected by the univariate Control Charts-based monitoring.

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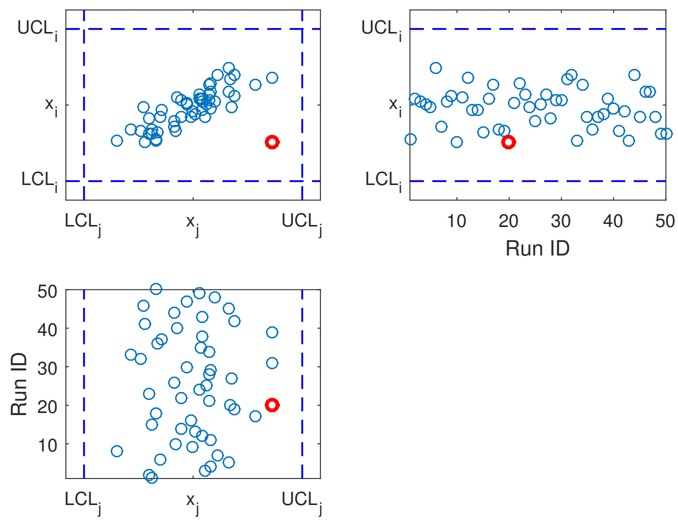

Fig. 1. Univariate Control Charts; in this 2-dimensional example, an outlier, represented by the red circle, (Top-left panel) cannot be detected by monitoring the control charts related to x_i (Top-right panel) and x_j (Bottom-left panel).

Nowadays, with the increasing availability of data in production and equipment, the need for multi-variate, high-dimensional *Anomaly Detection/Isolation* techniques is thriving; instruments to implement efficient AMSs are provided by Machine Learning (ML). ML approaches have proliferated in recent years Advanced Process Control (APC) solutions for Semiconductor Manufacturing [9], [10], thanks to the algorithmic advancements in the field and the increased computational and storage capabilities in the IT architecture of the Fabs; ML-based approaches have been used for Virtual Metrology, Predictive Maintenance and Fault Detection applications.

In this work we employ a state-of-the-art ML approach, *Isolation Forests (iForests)*, to detect outliers/anomalies in a high-dimensional monitoring problem; a iForest-based strategy is compared to other monitoring strategies on a real industrial dataset related to a Semiconductor Manufacturing Etching process. The contributions of the paper are: (i) proposing an i-Forest-based strategy to Anomaly Detection (AD) and (ii) apply such approach in a real industrial Semiconductor Manufacturing application.

The paper is organized as follows: Section II is devoted to present iForest, while in Section III the i-Forest AD

Symbol	Meaning
N	Observations available in the dataset
p	Number of monitored features
q	A generic feature
$X = \{x_1, \dots, x_n\}$	Input matrix ($x_i \in \mathbb{R}^p$)
x_O	An outlier
x_{new}	A new observation, to be monitored
s	Anomaly Score (AS)
τ	Threshold on AS
t	Number of binary trees
$X' \subset X \vert \Psi = \vert X' \vert$	A random subset of with the Input matrix and its cardinality
K_{INNER}	Monte Carlo Simulations (Inner cycle)
K_{OUTER}	Monte Carlo Simulations (Outer cycle)

TABLE I
SYMBOLS EMPLOYED IN THIS PAPER.

strategy is depicted. In Section IV the Semiconductor Manufacturing case study is illustrated and the Experimental results are provided; finally, in Section V the concluding remarks are provided.

A list of the notation employed in this work is reported in Table I.

II. Isolation Forest

A. IForest Concept

Isolation Forest (iForest) [11] is an AD method inspired by Random Forest [12], a well-known algorithm for regression and classification; iForest has been shown to outperform state-of-the-art outlier detection approaches in several applications (for example biomedical, ecological and traffic-related [13]; iForest is an extension of decision trees based on a mechanism called isolation: a procedure that through iterative partitioning of the input space aims to separate a new observation from the rest of the data at hand.

The main idea of iForest is that anomalies are few in number and much different from the rest of the data at hand and therefore are susceptible to the isolation procedure; therefore, it is reasonable to assume that such outliers will be divided from the rest of the data through simple partitioning. This procedure is illustrated in Figure 2, where an outlier x_O is isolated through an iterative procedure that required 3 partitions, while an inlier x_i is isolated by a procedure that required 9 partitions.

The isolation procedure generates a tree where at each leaf there is an observation and each internal node is associated with a split on one variable: the split value for each node is chosen randomly on a different subset of the data at hand. The iForest is an ensemble approach [14]: the aforementioned isolation procedure is repeated t different times generating (given the randomness of the approach) different trees; the indication of how likely an observation can be considered an outlier is provided by a score that is correlated on path lengths necessary to isolate that observation.

Besides accuracy, the iForest has several advantages over other AD methodologies:

- It does not require a model to describe the input-output relationship of the process to be monitored;
- It is computationally efficient w.r.t. common density or distance-based approaches to monitoring;
- Low memory requirement;
- Natural parallel computing implementation.

B. The iForest Algorithm

The random partitioning procedure can be represented by an ensemble of t binary trees. Anomalies produces mean paths (from root to leaves) which are longer w.r.t. normal attributes. Trees are called isolation trees (iTs).

Given a dataset $X = \{x_1, \dots, x_n\}$, $x \in \mathbb{R}^p$, each iT is obtained selecting a random subset $X' \subset X$ $(\psi = |X'|)$ of attributes and dividing X' by randomly selecting a feature q and a split value $\bar{q}$ until node has only one instance. It should be noticed that the characteristic of iTs enables to exploit subsampling, making this model capable to scale up to handle extremely large data size and high dimensional problems.

The iForest defines an *Anomaly Score (AS)* s, a quantitative index that defines the ”outlierness” degree of an observations; the AS is defined for an observation x by

$$s(x, \psi) = 2^{\left(-\frac{E(h(x))}{c(\psi)}\right)} \in [0, 1] \tag{1}$$

where $E(h(x))$ is the average path length $h(\cdot)$ over the t iTs and $c(\psi) = E(h(x)|\psi)$ is an adjustment factor that takes into account the cardinality of the subsampled dataset. When $E(h(x)) \to n - 1$, the AS tends to 0 meaning that x appears to be a normal instance. On the other hand when $E(h(x)) \to 0$, the AS $s \to 1$, meaning that x appears to be an outlier.

One of the most important advantages of iForest is that no distances or density measures are employed to detect anomalies: this eliminates a major computational cost of distance calculation in all distance-based and density-based methods. The parameters to tune in iForest are:

- the subsample size ψ which can remain relatively small;
- the number of trees t, which should be enough to allow path length convergence.

In this work, following the guidelines provided by the authors in [11], we have employed $\psi = 256$ and $t = 100$.

III. Anomaly Detection

The AD procedure adopted in this work is illustrated in Fig. 3; as represented, the AD module is composed by two parts:

- the *off-line* part that is dedicated to define/update the anomaly detection, evaluate the performances of the AD solution and choosing how to activate flags/alerts based on the AD score;

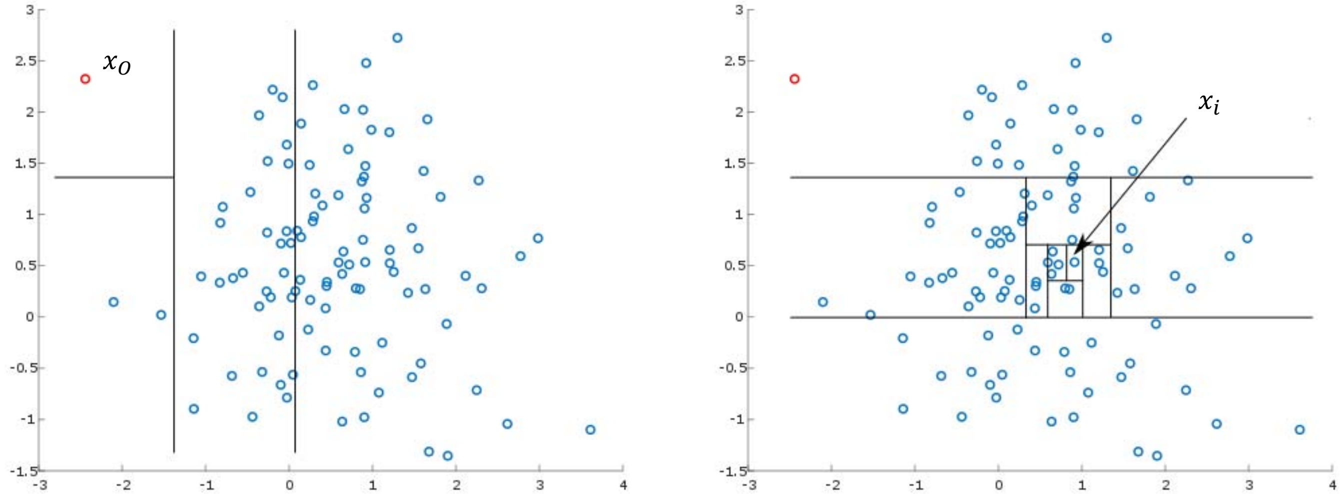

Fig. 2. The iForest procedure in case of an outlier (left panel) and an inlier (right panel): it can be appreciated how the iForest procedure took 3 partitions for isolating the outlier x_O and 9 partitions for isolating the inlier x_i.

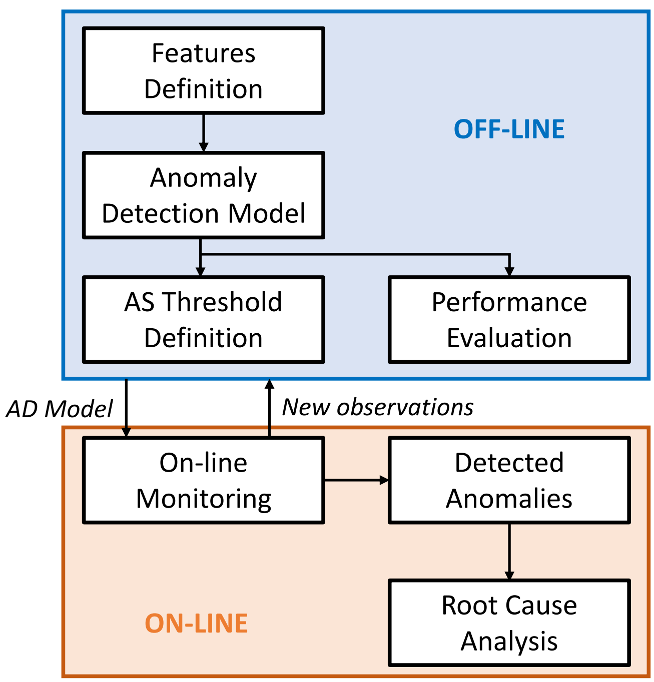

Fig. 3. Scheme of ML-based AD scheme.

- the *on-line* part that is devoted to evaluate the AD score for the observation at hand, highlight the presence of anomalies and, typically, perform the Root Cause Analysis, that is the process of pinpoint the causes of an abnormal behavior.

IForest, like many others ML-based AD technologies (for example Local Outlier Factor [15] or Online Over-sampling Principal Component Analysis [16]), defines an Anomaly score (AS) (for the iForest see eq. 1); Automatic Monitoring policies are generally based on triggering a reaction/intervention if the AS computed for a new observation overstep a predefined threshold τ.

How such threshold τ is defined? In this work we adopt a cross-validation (CV) approach to ensure that the chosen τ is associated with an optimal trade-off of two errors:

- *type I error (false positive FP)*, normal runs that are detected as anomalies;
- *type II error (false negatives FN)*, anomalies not detected as such.

We employed here a Monte Carlo Cross-Validation (MCCV) procedure (also known as Repeated Random Sub-sampling Validation) [14]; the portion of data employed to construct the iForest and benchmark AD methodologies are divided into

- training data (67%), a set of observations on which the AD models are constructed;
- validation data (the remaining 33% of data), used for evaluating the performances of the compared methodologies for different values of τ.

The split between training and validation is done at random, preserving the ratio between outlier and inlier observations in each set (*stratified cross-validation*). The aforementioned procedure is repeated $K_{\text{INNER}} = 100$ times, by randomly choosing how data are assigned to the training or validation set: results are reported in the following as average over the K MCCV simulations. The 'optimal' value of τ is then chosen as the average value over the K_{INNER} repetitions; once this has been set, the final model is computed over the combined training and validation data. Three final remarks:

- this procedure is called *inner* CV to differentiate it from the 'outer' CV procedure (detailed in Section IV) for comparing different AD approaches;

- in this work FP and FN errors has been considered as having the same associated costs; the optimization of τ is therefore done in order to minimize the total number of errors;
- in some industrial settings the tuning of τ can be performed on-line, in the sense that users can decide to adapt the threshold dynamically during the production.

IV. Etching: Experimental Results

A. Dataset Description

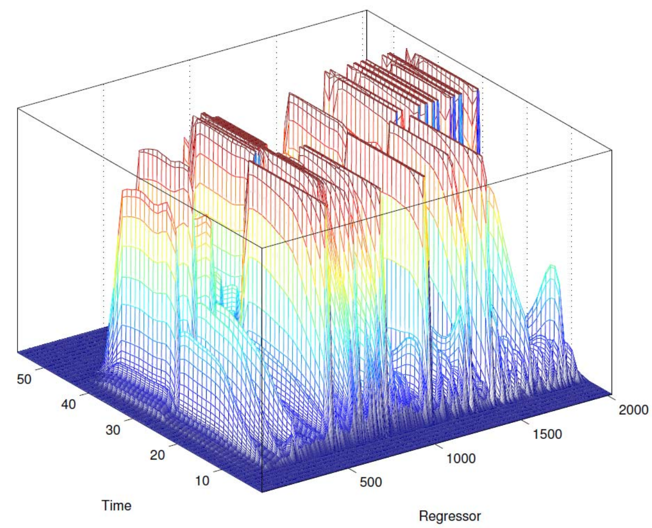

Fig. 4. Optical Emission Spectroscopy data associated with an Etching run.

The iForest-based AD strategy have been tested on a real industrial dataset related to a Semiconductor Manufacturing Etching process [17].

The data available consists of 2194 wafers belonging to a set of 313 (not complete) lots, for which Optical Emission Spectrometry (OES) are available. The process in question is plasma etching, for which OES data represents a non-costly and informative source of information from a chemical point of view. The dataset also includes information on another important quantity, the Etching Rate that is the ratio between the depth of the created trench and the time taken to perform the excavation.

One of the main issue in studying AD problems is the availability of tagged data, where anomalies are indicated; in this work we overcome this issue by employing the Etch Rate as a quality indicator for the produced wafers: outliers or changes seen in the Etch Rate can be considered as anomalies or changes in the production. Unfortunately, the Etch Rate is costly to compute in production and in some production settings is not available for most produced wafers; this scenario underlines the importance of an AMS that is able to infer anomalous situations from different data sources, like OES data [18]. The available dataset consists of 2170 inlier wafers and 24 outlier wafers, labeled as such by inspecting the Etch Rate. From such data, descriptive features are extracted to be employed in the iForest and the compared AD procedures.

B. Preprocessing

The OES data can be monitored on-line during the wafer processing, but OES data cannot be directly fed into the iForest in their raw form, since they cannot be formatted into a design matrix $X \in \mathbb{R}^{N \times p}$; a preprocessing step, called *Feature Extraction*, need to be performed in order to enable the anomaly detection training and monitoring.

The design of features aim to characterize the majority of the informative content contained in the underlying data by an automatic procedure. A set of $p = 652$ features is here defined by computing mean, max, min and variance[1] for a predefined set of 163 wavelengths.

C. Experimental Settings

IForest has been compared with a Control Chart based-approach to Anomaly Monitoring and with another state-of-the-art anomaly detection approach for high-dimensional data, called Angle-Based Outlier Detection (ABOD) [19]. Angle Based Outlier Detection (ABOD) is an angle-based method, where angles are computed between an observation in exam and other samples available in the dataset. The intuition behind ABOD, is that inliers will produce angles with high variance since they are inside a cluster, while outliers will have associated angles with low variances since they are outside a cluster. Dedicated AS for Control Charts and ABOD are computed by performing an Inner CV procedure as described in Section III.

The proposed methodologies have been compared through a second MCCV) procedure, called *Outer CV*; the $N = 2194$ have been divided into:

- training and validation data (50%), a set of observations on which the *Inner CV* is performed for constructing the AD models and the thresholds τ ASs (as detailed in Section III;
- test data (the remaining 50% of data), used for evaluating the performances of the compared methodologies.

The split between training and validation is done at random, preserving also in this case the ratio between outlier and inlier observations in each set. The aforementioned procedure is repeated $K_{\text{OUTER}} = 100$ times: results are reported in the following as average over the K_{OUTER} MCCV simulations.

[1]The aforementioned statistics are computed on a central, informative part of the Etching process: the OES are windowed over time.

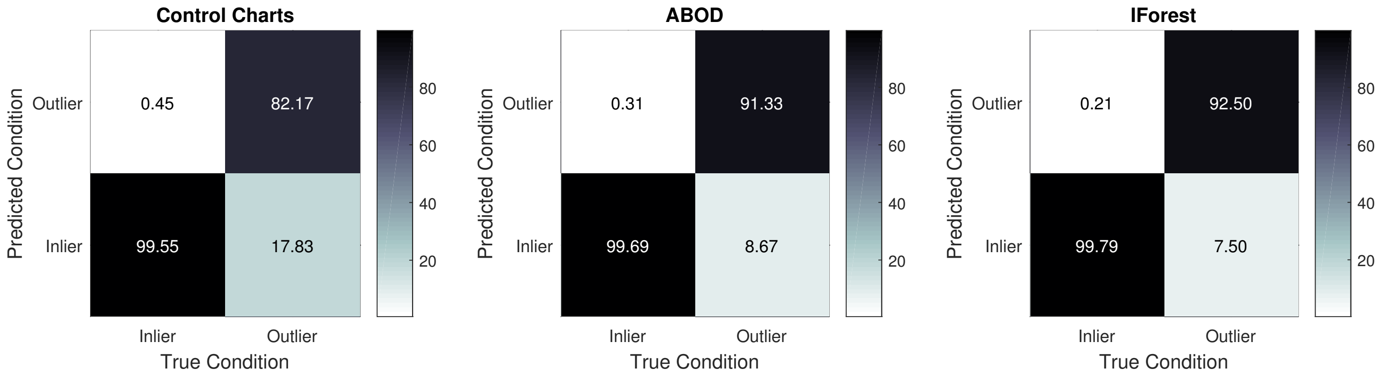

Fig. 5. Confusion matrices of the compared methodologies.

Method	**Precision**	**Recal**
Control Charts	66.94%	82.17%
ABOD	76.32%	91.33%
iForest	**82.84%**	**92.50%**

TABLE II
PERFORMANCES OF THE CONSIDERED AD APPROACHES WITH THE ETCHING DATASET.

D. Results

The experimental results are reported in Fig. 5 in the form of *confusion matrices* (also called error matrices), where each row of the matrices represents the instances in a predicted class (inlier or outlier) while each column represents the instances in an actual class. It can be appreciated how the Control Chart AD-based approach is outperformed by both multi-dimensional AD approaches in terms of both false negatives and false positives. Moreover, iForest outperfom ABOD in each type of error.

The experimental results are reported in Table 2 in terms of *precision* and *recall*; the two indicators are defined as:

$$\text{precision} = \frac{TP}{TP+FP},$$
$$\text{recall} = \frac{TP}{TP+FN},$$

where TP and TN are, respectively, the amount of True Positives (outlier correctly predicted as such) and True Negatives (inlier correctly predicted as such). From Table 2 it can be appreciated how the multi-dimensional AD methods guarantee above 76% precision and above 91% recall. Moreover, it can be appreciated how the iForest outperforms significantly ABOD in terms of precision.

V. CONCLUSIONS

In this paper, iForest has been employed for Anomaly/Outlier detection on a real industrial case study; the case study under examination was related to Etching, one of the main steps in the Semiconductor Manufacturing fabrication. IForests have been compared with classical univariate Control Charts-based approach to AD and with another multi-dimensional popular approach to AD, called ABOD. It has been shown how monitoring the OES data provided satisfying results in detecting outliers; on the dataset at hand the Control Charts-based approach to AD was outperformed by multi-dimensional approaches and iForests achieved the best performances in terms of recall and precision.

Future works will regard more sophisticated/automatic approaches to extract features from OES data; moreover, new policies for setting up the threshold τ on the anomaly score will be investigated. Furthermore, extended comparison with other multi-dimensional AD methodologies, like onlinePCA, osPCA [16] and LOF [15], will be performed.

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